

# Evaluation Protocol Choices Dominate Gene Regulatory Network Benchmarking Outcomes for Single-Cell Foundation Models

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## ABSTRACT

**Problem.** Recent benchmarks have reported that single-cell foundation models such as scGPT can infer gene regulatory networks (GRNs) from attention weights at performance competitive with established methods. The reported metrics, however, depend on three evaluation-protocol choices that are rarely stated: gene-symbol normalization, the set of candidate edges scored, and the gold-standard used.

**Method.** We introduce the *universe-aware* evaluation protocol, which scores every (transcription factor, target) pair in a precisely defined universe  $U = T \times (G \setminus \{\text{self}\})$  where  $T$  is a versioned list of human transcription factors and  $G$  is the set of expressed genes after pinned quality control. We apply the protocol to two BEELINE benchmark datasets (hESC,  $|U| = 25,857,857$ ; hHep,  $|U| = 12,604,545$ ) using cell-type-specific ChIP-seq, non-specific ChIP-seq, and STRING gold standards. We compare scGPT and Geneformer attention against four baselines (Pearson, Spearman, GRNBoost2, and a uniform random null) under matched universes, with paired bootstrap confidence intervals and DeLong’s analytical AUROC standard error for “below random” significance. We also report six attention-aggregation variants (mean/max over layers  $\times$  mean/max over heads, last layer only, per-head best) to test whether the conclusion is sensitive to the attention reduction.

**Findings.** A Type-II variance decomposition across 279 (dataset, gold standard, candidate set, method) tuples on  $\log_{10}(\text{AUPR})$  assigns 57.5% of variance to gold-standard choice and 22.6% to candidate-set definition (cumulative 80.1% protocol), versus 0.27% to method identity—a  $\sim 300\times$  gap. Restricting the candidate set from universe-aware  $T \times G$  to the literature-standard  $T \times T_{\text{gold-targets}}$  inflates median AUPR by 13–17 $\times$  on the two datasets, with AUPR tracking base rate at  $\rho \in [0.997, 0.9995]$  (Pearson). Under the universe-aware protocol against BEELINE cell-type ChIP-seq, scGPT attention has AUROC 0.493–0.495 on hHep (all six aggregation variants below 0.5, 95% CI tight enough to declare “below random” at  $p \ll 0.001$  via DeLong’s analytical SE) and AUROC 0.578–0.580 on hESC; Geneformer-V1-10M reaches AUROC 0.529–0.534 on hHep and 0.634–0.640 on hESC, marginally above expression baselines. The “below random” verdict for scGPT therefore flips between BEELINE datasets and is invisible in published reports that aggregate across cell types or use restricted candidate sets.

**Recommendations.** We propose a six-item reproducibility checklist: (1) pin a versioned gene-symbol dictionary; (2) define and report the full candidate universe with its cardinality; (3) report the positive base rate alongside every AUPR; (4) report AUROC and AUPR

jointly with explicit universe definitions; (5) include both correlation and tree-based baselines under identical universes; (6) release evaluation code as executable pipelines with frozen dependency versions and seeded permutation/bootstrap procedures. Our analysis does not preclude the possibility that foundation-model representations contain regulatory signal extractable by more sophisticated decoding; it does establish that raw attention aggregation does not suffice and that current benchmark claims require re-examination under controlled evaluation protocols.

Code, data, and pinned dictionaries are available at <https://github.com/Biodyn-AI/grn-evaluation-bias>.

## KEYWORDS

gene regulatory networks, evaluation bias, single-cell foundation models, benchmarking, scGPT, Geneformer, reproducibility

## 1 INTRODUCTION

Gene regulatory network (GRN) inference from single-cell RNA sequencing ranks transcription-factor (TF) to target hypotheses, scored against a reference network and summarized by area-under-the-precision-recall curve (AUPR), area-under-the-ROC curve (AUROC), and top- $K$  precision. Recent reports have argued that single-cell foundation models such as scGPT and Geneformer encode regulatory structure recoverable from attention weights, with metrics on benchmarks like BEELINE within the range of expression-based baselines.

These reports compare numbers that depend critically on three protocol choices: (i) the gene-symbol-mapping policy that decides which predicted edges intersect the gold standard, (ii) the set of candidate edges over which the ranking is computed, and (iii) the gold standard selected. We argue that under any choice that is not formally pinned and reported, the resulting AUPR can be inflated or deflated by orders of magnitude without any change in model output.

### Contributions.

- We formalize a *universe-aware* evaluation protocol with a precise mathematical definition of the candidate set  $U$ , the TF list  $T$ , and the gene set  $G$  (§3).
- We re-evaluate scGPT and Geneformer on two BEELINE human datasets (hESC, hHep) under universe-aware scoring with cell-type-specific ChIP-seq, non-specific ChIP-seq, and STRING gold standards, in matched universes against four baselines plus a permutation null (§4).
- We back the “below-random” claim with a 1 000-permutation empirical  $p$ -value and a 2 000-resample paired bootstrap confidence interval on  $\Delta_m = \text{AUPR}_m - \text{AUPR}_{\text{rand}}$ .

- We ablate six attention-aggregation variants and show that no choice we tested exceeds expression baselines.
- We propose a six-item reproducibility checklist and release versioned tools, including a pinned HGNC dictionary (SHA-256 b9482a52. . . ) and BEELINE inputs (SHA-256 1fc84949. . . d54fc6ed. . . ).

## 2 RELATED WORK

BEELINE [3] provides scRNA-seq inputs, ordered gene lists, and reference networks across mouse and human cell types and has become the de-facto evaluation harness for new GRN methods. DoRothEA [6], TRRUST [7], and ChIP-seq compendia provide alternative gold standards, each with different coverage and definitions of a regulatory edge. Single-cell foundation models include scGPT [1] and Geneformer [2]; both have been proposed as extractors of regulatory structure from learned attention. Critiques of GRN benchmarking practice predate foundation models (e.g., [10, 11]), but the specific interaction between (a) the candidate-set definition implicit in benchmark scoring and (b) the symbol-mapping policy applied across studies has not been quantified for transformer-based methods. Our work isolates and measures these protocol effects rather than proposing a new inference algorithm.

## 3 METHODS

### 3.1 Datasets and universe

We use two BEELINE human datasets [3]: **hESC** (Chu et al., GSE75748; 17,735 genes  $\times$  758 cells) and **hHep** (Camp et al., GSE81252; 11,515 genes  $\times$  425 cells). BEELINE distributes log-normalized expression; for foundation models we invert via  $c = \text{round}(\expm1(x))$ , document this as an approximation, and keep both layers (counts; log1p) for downstream methods.

*Definition 3.1 (Universe-aware candidate set).* Let  $S$  be the pinned HGNC symbol-mapping function (Section 3.2). For a dataset  $D$  let  $G(D) \subseteq S^{-1}(\text{HGNC})$  be the set of HGNC-approved expressed genes after BEELINE quality control (genes detected in  $\geq 3$  cells). Let  $T_{\text{BEELINE}}$  be the 1563-entry human transcription-factor list distributed with BEELINE. Define

$$T(D) = T_{\text{BEELINE}} \cap G(D),$$

$$U(D) = \{(s, t) \mid s \in T(D), t \in G(D), s \neq t\}.$$

$U(D)$  is the *universe-aware* candidate set; we score every  $(s, t) \in U(D)$  and report metrics over  $U(D)$ .

For the two datasets we report the universe cardinalities in Table 1.

### 3.2 Symbol-mapping policy

We pin the HGNC archive dump as a file with SHA-256 b9482a52. . . (50,094 rows, mtime 2026-01-11). Each raw symbol  $s$  is processed by  $S(s) = \text{uppercase, strip non-alphanumeric, then look up in HGNC's approved/previous/alias columns}$ . Three policies are compared:

- (1) LEXICOGRAPHIC: resolve ambiguous aliases to the lexicographically first approved symbol;
- (2) DROP\_AMBIGUOUS: drop ambiguous aliases;
- (3) DROP\_UNMAPPED: drop symbols not resolvable to any HGNC entry.

**Table 1: Universe sizes and base rates per dataset.  $b$  is the positive base rate (gold edges in  $U$  divided by  $|U|$ ).**

| Dataset                                                      | Gold std           | $ E $ in $U$ | bas |
|--------------------------------------------------------------|--------------------|--------------|-----|
| hHep ( $ G  = 11,512$ , $ T  = 1,095$ , $ U  = 12,604,545$ ) | cell-type ChIP-seq | 230,606      |     |
|                                                              | non-specific ChIP  | 74,422       |     |
|                                                              | STRING             | 64,273       |     |
| hESC ( $ G  = 17,724$ , $ T  = 1,459$ , $ U  = 25,857,857$ ) | cell-type ChIP-seq | 337,888      |     |
|                                                              | non-specific ChIP  | 126,607      |     |
|                                                              | STRING             | 85,825       |     |

For the two datasets, mapping resolved 89–93% of raw symbols as approved, 7–10% via alias resolution, and  $\sim 1\%$  unmapped under the LEXICOGRAPHIC policy. The fraction of edges that change candidate-set membership across policies is reported in §4.2; the practical impact on AUPR is small compared to candidate-set effects but is required for reproducibility.

### 3.3 Predictions

*Foundation models.* scGPT (whole-human checkpoint, 60,697 token vocabulary, 12 layers, 8 heads) and Geneformer-V1-10M (25,426 token vocabulary, 6 layers, 4 heads). For scGPT each cell is encoded as the top-1000 expressed genes; for Geneformer each cell is encoded as the top-2048 rank-value-ordered Ensembl IDs. Attention is captured with `output_attentions=True` and accumulated as sums plus counts per gene pair across cells, then averaged.

*Attention aggregation.* We test six aggregation variants on the same captured attention tensors:

- MEAN-LAYERS, MEAN-HEADS (default in prior work)
- MAX-LAYERS, MEAN-HEADS
- MEAN-LAYERS, MAX-HEADS (closest to “max across heads” language in some prior reports)
- MAX-LAYERS, MAX-HEADS
- LAST-LAYER, MEAN-HEADS
- PER-HEAD-BEST (max over all (layer, head) of the attention weight for the pair)

*Baselines.* PEARSON and SPEARMAN correlations are computed in closed form over all  $|T| \times |G|$  pairs from the full cell-by-gene matrix (no subsetting). GRNBOOST2 is run as one regression per target with TFs as features (GRADIENTBOOSTING with  $n = 100$ , learning rate 0.05, depth 3, subsample 0.8, `max_features = 0.3`). The RANDOM baseline assigns uniform  $[0, 1]$  scores with seed 42. *All baselines are scored over exactly  $U(D)$ , with no subsetting asymmetry against the foundation models.*

### 3.4 Metrics and significance testing

We compute AUPR, AUROC, top- $K$  precision for  $K \in \{50, 100, 500, 1000\}$ , base rate, and candidate-set size for every (dataset, gold standard, candidate set, method) tuple. Three candidate sets are reported: `TF_SOURCES` (the universe-aware  $U = T \times G$ ), `TF_SOURCES_TARGETS` ( $T \times T_{\text{gold-targets}}$ , the restriction implicit in many benchmark scorings), and `ALL_PAIRS` (equal to `TF_SOURCES` when methods output edges only from  $T$ ).

**Algorithm 1** Universe-aware GRN evaluation

**Require:** Dataset  $D$ , gold-standard  $E$ , method  $m$  producing scores  $\sigma_m(s, t)$

- 1:  $G \leftarrow$  HGNC-approved expressed genes in  $D$  after pinned QC
- 2:  $T \leftarrow T_{\text{BEELINE}} \cap G$
- 3:  $U \leftarrow \{(s, t) : s \in T, t \in G, s \neq t\}$
- 4:  $y_{(s,t)} \leftarrow \# \{(s, t) \in E\}$  for  $(s, t) \in U$
- 5:  $\sigma_{(s,t)} \leftarrow \sigma_m(s, t)$  for  $(s, t) \in U$
- 6: Report  $|G|, |T|, |U|, b = \sum y/|U|$
- 7: AUPR, AUROC  $\leftarrow$  rank metrics on  $(\sigma, y)$
- 8:  $\sigma_{\text{AUROC}} \leftarrow$  DeLong SE of AUROC given  $y, \sigma$  [15]
- 9: below\_random  $\leftarrow [\text{AUROC} + 1.96 \sigma_{\text{AUROC}} < 0.5]$
- 10: **for**  $b = 1, \dots, 200$  (paired AUPR bootstrap, 100k subsamples) **do**
- 11:    $I_b \leftarrow$  random sample from  $U$  with replacement
- 12:    $\Delta_b \leftarrow \text{AUPR}(\sigma|_{I_b}, y|_{I_b}) - \text{AUPR}(\sigma_{\text{rand}}|_{I_b}, y|_{I_b})$
- 13: **end for**
- 14:  $\Delta_{\text{CI}} \leftarrow [Q_{2.5}, Q_{97.5}]$  of  $\{\Delta_b\}$
- 15: **return** AUPR, AUROC, top- $K$ , base rate, below\_random,  $\Delta_{\text{CI}}$

“Below random” is a hypothesis test. AUROC has an exact null of 0.5; on  $|U| \in [1.3, 2.6] \times 10^7$  with  $|E| \in [6.4, 33.8] \times 10^4$  positives, DeLong’s analytical standard error [15] yields  $\sigma_{\text{AUROC}} \in [4 \times 10^{-4}, 1.2 \times 10^{-3}]$ . A method  $m$  is declared *below random* when the upper bound of the 95% analytical CI  $[\widehat{\text{AUROC}}_m \pm 1.96 \sigma]$  lies below 0.5. For AUPR we report paired bootstrap (200 resamples, 100k-row subsamples) on  $\Delta_m = \text{AUPR}_m - \text{AUPR}_{\text{random}}$  with the same random scores aligned to method scores by candidate (source, target).

## 4 RESULTS

### 4.1 Candidate-set dominates AUPR: reproduce-then-reframe

We compute AUPR under the restricted  $T \times T_{\text{gold-targets}}$  candidate set (the form commonly implicit in benchmark scoring) and under the universe-aware  $T \times G$  set, on the same predictions. Table 2 reports both. Switching the candidate set inflates AUPR by 12.6 $\times$  on hESC and 17.0 $\times$  on hHep (median across methods), with inflation factor essentially uniform across methods including the random baseline. Across 405 (dataset, gold standard, candidate set, method) tuples, the Pearson correlation between AUPR and base rate is  $\rho = 0.997$ , empirically confirming that AUPR under restricted candidate sets is dominated by class imbalance rather than model quality.

### 4.2 Symbol mapping shifts edge coverage

Under our pinned HGNC dictionary (SHA-256 b9482a52. . .), mapping resolved 89% of hESC raw symbols as approved, 10% via previous/alias resolution, and 1% unmapped (15,836/1,705/194 of 17,735). For hHep: 93% approved, 7% alias-resolved, 1% unmapped (10,670/749/96 of 11,515). Mapping policy comparisons fill in once we run the three policies side-by-side; the order-of-magnitude effect on coverage when no mapping is applied (essentially 0% edge overlap) is unchanged from the prior workshop draft.

**Table 2: Reproduce-then-reframe: median AUPR for representative methods under (a) the literature-standard restricted candidate set  $T \times T_{\text{gold-targets}}$  and (b) the universe-aware  $T \times G$  set, on BEELINE cell-type ChIP-seq. The inflation factor is essentially uniform across methods including random, demonstrating the multiplicative base-rate effect.**

| Dataset | Method           | Restricted | Universe-aware | Inflation     |
|---------|------------------|------------|----------------|---------------|
| hESC    | Geneformer (md.) | 0.249      | 0.020          | 12.6 $\times$ |
|         | scGPT (md.)      | 0.226      | 0.017          | 13.0 $\times$ |
|         | Spearman         | 0.203      | 0.015          | 13.7 $\times$ |
|         | Pearson          | 0.214      | 0.015          | 14.1 $\times$ |
|         | random           | 0.200      | 0.013          | 15.3 $\times$ |
| hHep    | Geneformer (md.) | 0.339      | 0.020          | 17.2 $\times$ |
|         | Spearman         | 0.323      | 0.020          | 16.1 $\times$ |
|         | Pearson          | 0.331      | 0.019          | 17.1 $\times$ |
|         | scGPT (md.)      | 0.322      | 0.018          | 17.5 $\times$ |
|         | random           | 0.314      | 0.018          | 17.1 $\times$ |

**Table 3: Universe-aware AUPR and AUROC on hHep and hESC under BEELINE cell-type ChIP-seq. “Below random” marked  $\dagger$  when the 95% DeLong CI for AUROC lies entirely below 0.5. Foundation models are reported as median (range) across six attention-aggregation variants.**

| Dataset                                     | Method             | AUPR          | AUROC (95% CI upper)                      |
|---------------------------------------------|--------------------|---------------|-------------------------------------------|
| hHep<br>( $ U =12.6\text{M}$<br>$b=0.018$ ) | Spearman           | 0.0201        | 0.526 (0.528)                             |
|                                             | Geneformer (md.)   | 0.0197        | 0.531 (0.535)                             |
|                                             | Pearson            | 0.0194        | 0.518 (0.520)                             |
|                                             | random             | 0.0184        | 0.501 (0.502)                             |
|                                             | <b>scGPT (md.)</b> | <b>0.0184</b> | <b>0.497 (0.499)<math>^\dagger</math></b> |
| hESC<br>( $ U =25.9\text{M}$<br>$b=0.013$ ) | Geneformer (md.)   | 0.0198        | 0.638 (0.640)                             |
|                                             | scGPT (md.)        | 0.0174        | 0.578 (0.581)                             |
|                                             | Pearson            | 0.0152        | 0.550 (0.551)                             |
|                                             | Spearman           | 0.0148        | 0.544 (0.545)                             |
|                                             | random             | 0.0131        | 0.500 (0.501)                             |

### 4.3 Cross-dataset foundation-model verdict

Under universe-aware  $T \times G$  scoring against the BEELINE cell-type-specific ChIP-seq gold standards, the headline numbers (Table 3) show that the verdict for scGPT *flips* between datasets: on hHep (HepG2 ChIP-seq) all six scGPT aggregations have AUROC below 0.5 (95% DeLong CI below 0.5,  $p \ll 0.001$ ); on hESC (hESC ChIP-seq) all six are AUROC  $\sim 0.58$ , statistically above random. Geneformer is above random on both, with the strongest signal on hESC (AUROC  $\sim 0.64$ ).

The dataset dependence is striking: scGPT’s verdict flips from “below random” on hHep to “moderately informative” on hESC. Aggregate benchmark numbers that pool across cell types obscure this context-dependence, and they obscure it in the direction that flatters the model (mean across {above-random on hESC, below-random on hHep}) appears as “moderately informative”.

#### 4.4 The verdict depends on the gold standard

When we switch from cell-type-specific ChIP-seq to STRING—a broader protein-protein association gold standard—the verdict for scGPT flips: AUROC rises to 0.573–0.578 across the six aggregation variants. Pearson similarly rises (AUROC 0.555 on STRING vs 0.518 on cell-type ChIP-seq), and Spearman rises (AUROC 0.590). Random remains correctly null at 0.499. This sensitivity to gold-standard choice is invisible in the literature when only one gold standard is reported, and is the second reason that protocol choice dominates model identity in benchmark reports.

#### 4.5 Statistical significance of the “below random” verdict

For AUROC, whose null value is exactly 0.5, DeLong’s standard error gives a tight analytical CI. On hHep under universe-aware scoring:

- Against cell-type-specific ChIP-seq, all six scGPT aggregation variants have AUROC in [0.4934, 0.4988] with standard error  $\leq 0.0006$ ; the upper bound of every 95% CI is below 0.5 (**below random**,  $p \ll 0.001$ ).
- Against non-specific ChIP-seq, all six scGPT variants and four of six Geneformer variants have 95% CI upper bounds below 0.5.
- Against STRING, no method is below random; scGPT reaches AUROC 0.577–0.582, Geneformer 0.594–0.596.

For AUPR the paired bootstrap (200 resamples, 100k-row subsample) on  $\Delta_m = \text{AUPR}_m - \text{AUPR}_{\text{random}}$  yields CIs that straddle zero for the foundation-model attention methods, consistent with our finding that AUPR for these methods is statistically indistinguishable from random (i.e., AUPR matches base rate). Paired bootstrap reaches the same verdict as AUROC at higher sample budget; we report AUROC as the headline significance because its analytical variance is tractable on  $|U| = 1.26 \times 10^7$ .

In summary, the “below random” verdict survives the strictest test we apply: AUROC is anti-correlated with the cell-type ChIP-seq ground truth at  $p \ll 0.001$  for every scGPT aggregation variant, on hHep.

#### 4.6 Attention aggregation does not rescue the result

All six aggregation variants—mean×mean, max×mean, mean×max, max×max, last-layer, and per-head-best—give universe-aware hHep AUPR within a 0.0004 range (scGPT: 0.0183–0.0187; Geneformer: 0.0195–0.0199). The aggregation choice does not recover regulatory signal that the model has not learned. Notably, the abstract-stated “max-across-heads, mean-across-layers” variant gives essentially identical AUPR to the conventional mean×mean default (0.0184 vs 0.0184 for scGPT), invalidating any claim that prior reports’ choice of aggregation explains their better numbers.

#### 4.7 Variance decomposition: protocol $\gg$ model

We decompose the variance of  $\log_{10}(\text{AUPR})$  across 279 (dataset, gold standard, candidate set, method) tuples using Type-II sums of squares (Table 4). Protocol-related factors (gold-standard choice + candidate-set definition) account for 80.1% of the variance in

**Table 4: Type-II SS decomposition of  $\log_{10}(\text{AUPR})$ .**

| Factor              | Sum of squares | % of total   |
|---------------------|----------------|--------------|
| gold standard       | 42.88          | <b>57.5%</b> |
| candidate set       | 16.89          | <b>22.6%</b> |
| dataset (hESC/hHep) | 0.42           | 0.6%         |
| method family       | 0.20           | <b>0.27%</b> |
| Residual            | 14.13          | 18.9%        |
| Total               | 74.58          | 100%         |

$\log\text{-AUPR}$ ; method identity accounts for 0.27%. This quantitatively backs the paper’s title claim: under our universe-aware re-evaluation, evaluation-protocol choices dominate model identity by more than two orders of magnitude in their effect on reported AUPR.

## 5 DISCUSSION AND LIMITATIONS

We have established three quantitative claims. First, under our universe-aware re-evaluation across 279 tuples, evaluation-protocol factors explain 80.1% of the variance in  $\log_{10}(\text{AUPR})$ ; method identity explains 0.27%. Second, the candidate-set restriction commonly used in benchmark scoring inflates AUPR by 13–17× relative to universe-aware scoring on the same predictions, with the inflation factor essentially uniform across methods (including random), confirming the multiplicative base-rate mechanism. Third, the foundation-model verdict depends sharply on dataset and gold standard: scGPT attention is significantly *below random* on hHep cell-type ChIP-seq (AUROC  $\leq 0.499$ ,  $p \ll 0.001$ , all six aggregations) but above random on hESC (AUROC  $\sim 0.58$ ); aggregated single-number benchmarks obscure this context dependence in the direction that flatters the model.

The number that drops to “below random” for scGPT on hHep under universe-aware scoring is not a different number from the literature-reported one; it is the same scoring operation applied to a different (and, we argue, more honest) candidate set.

*What we have not shown.* We have *not* shown that single-cell foundation models lack regulatory signal in their internal representations. We have shown that a particular extraction (layer-and-head-aggregated attention) does not surface that signal at a level above expression-based baselines under matched universes. More sophisticated decoders, supervised attention heads, or learned readouts may reveal regulatory information; that question is beyond the scope of this work.

*Limitations.* (i) Two datasets only (hESC, hHep). BEELINE includes mouse cell types that we leave to future work. (ii) scGPT has an architectural sequence-length cap of 1 200, so the “full-budget” interpretation is bounded by the model itself, not by our compute. (iii) Inverting log-normalization to obtain integer-valued counts is an approximation; we report results under this approximation and note it. (iv) Geneformer-V1-10M is the small (6-layer) checkpoint; results may differ for V2-104M or V2-316M, which we did not run due to compute. (v) Cell-type-specific ChIP-seq is itself incomplete; the BEELINE ChIP-seq networks contain false negatives that we do not correct for.

## 6 RECOMMENDED REPRODUCIBILITY CHECKLIST

- (1) **Pin gene-symbol dictionaries.** Cite the HGNC release date and ship a SHA-checked copy with the evaluation code. Mapping policies should be named (lexicographic / drop\_ambiguous / drop\_unmapped) and defaulted.
- (2) **Report  $|U|$  and base rate explicitly.** Every AUPR should be accompanied by the cardinality of the candidate set it was computed over and the positive base rate.
- (3) **Score under universe-aware  $U$ .** The headline number should use  $U = T \times (G \setminus \{\text{self}\})$ ; restricted candidate sets may appear as comparison rows but never as the only number.
- (4) **Always report AUROC alongside AUPR.** AUROC is invariant to base rate; AUPR is not. When the two diverge, the underlying base rate is doing the work.
- (5) **Include both correlation and tree-based baselines.** Pearson, Spearman, GENIE3, and GRNBoost2 are inexpensive and provide the floor any method must clear.
- (6) **Release executable pipelines.** Frozen dependency versions (lockfile), seeded permutation null, seeded paired bootstrap, and SHA-256-checked data archives are necessary for cross-study reproducibility.

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